Ex No: 07 Implementation of flow control at data
14.09.26 link layer using sliding window

Aim:

To write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCAL.

Code:-

SENDER.PY

```
import time
w = int(input("Enter window size: "))
mes = input()
frame = [[i, ch] for i, ch in enumerate(mes)]
base = 0
while base < len(frames):
    with open("sender_buffer.txt", "w") as f:
        for i in range(base, min(base + w,
                                 len(frames))):
            print(format(i))
    time.sleep(3)
    try:
        with open("receiver_buffer.txt", 'r').
            data = f.read().strip()
        if ack == base + 1:
            print("ACK recived: "ack)
            base = ack + 1
        else:
            print("Wrong")
```

```
elif data.startswith("NPCK"):
    print("NPCK Received Resending...")

except:
    print("waiting for Receiver...")
```

RECEIVER.PY

```
exp = 0
with open("sender_buffer.txt", "r") as f:
    lines = f.readlines()

ok = True
for line in lines:
    no, data = line.strip().split(":")
    if int(no) != exp:
        ok = False
        nack = exp
        break
    exp += 1

with open("Receiver_Buffer.txt", "w") as f:
    if ok:
        f.write(f"ACK:{exp-1}")
        print("ACK sent:", exp-1)
    else:
        print("NPCK sent:", nack)
```

Input:

Enter Window Size: 3

Enter Menage: HELLO

Output: (Sender)

Frames Sent:

[0, 'H']

[1, 'E']

[2, 'L']

Ack received: 2

Frames sent:

[3, 'L']

[4, 'O']

Ack received

(Receiver)

Ack sent: 2

Ack sent: 4

Result:-

Thus the program to implement flow control at data link layer using sliding window has been executed successfully.